

NEO Mathematics Supporting Adult Guidance

Pythagoras and the Hidden Length - 10-session unit

Connection before curriculum. Always.

The learner may arrive with disrupted mathematical experiences, anxiety about calculation, uneven prerequisite knowledge or strong mathematical curiosity alongside low confidence. The role of the Supporting Adult is not to accelerate the learner through every panel. Help the learner find an entry point, protect dignity, and use the lesson's explicit instructions to reduce uncertainty.

Stable shell, variable mathematical journey

Every session uses the same recognisable shell: Driving Question, Stepping Stones, six NEO Cornerstone panels, Scratchpad, Practice Companion/Studio and Final Task. The mathematical sequence varies according to conceptual dependency. Predictability of navigation is intentional, particularly for learners who benefit from reduced executive-function and interpretation demands.

Adult facilitation pattern

- Ask the learner to read the 'What to do' steps aloud or silently. Avoid replacing them with a longer verbal explanation unless requested.
- If the learner is unsure, ask: 'What is the next visible action?' rather than 'Do you understand?'
- Before calculations, return to geometry: Where is the right angle? Which side is opposite it? What role does the unknown side play?
- Use the Socratic prompt before a hint. Use the worked solution as a re-entry tool, not a punishment or last resort.
- Permit paper, voice, sketch, Scratchpad or another communication route where the mathematical evidence remains clear.
- Normalise pauses. Rest reduces load without reducing mathematical ambition.

Session-by-session mathematical purpose

1. The Harmony of Numbers

Do not turn the film into a comprehension test. The pre-viewing prompt is deliberately light. The goal is attraction to connection and pattern.

2. Three squares

Protect the area meaning of $a^2+b^2=c^2$. The formula is the compressed endpoint, not the starting point.

3. Hidden length

Method choice must follow side role. A common misconception is 'Pythagoras means add'; diagnose the hypotenuse before calculator work.

4. Practice Studio

The 16 questions are a 45-minute progression. A planned pause after Question 8 is part of the design. Do not use speed as the main success measure.

5. Measurement and modelling

Insist on the question 'Where is the right angle?' Real-life wording does not automatically justify Pythagoras.

6. Proof and Perigal

The drag-and-drop proof uses pieces that translate without stretching. Draw attention to the invariant: area. Compare finite testing with general proof.

7. Hippiasus and $\sqrt{2}$

The contradiction proof establishes irrationality; the dramatic details of Hippiasus's fate are historically uncertain. Preserve the lesson's distinction between mathematical certainty and historical evidence.

8. Triple families

Scaling is the core. Euclid's generating formula is a high-ceiling reveal, not a memorisation requirement.

9. Other shapes

The deeper structure is $A = k s^2$ for the same family of similar figures. Counterexamples are essential: fixed-height rectangles and mixed shapes clarify the condition.

10. Evidence Studio

Help the learner select evidence rather than copy every lesson. The final product should communicate Harmony, Relationship, Proof, Disruption and Utility.

Cornerstone interpretation in this unit

Connection reveals relationships, human story and mathematical dependency. **Movement** requires something the learner can change and a mathematical response they can observe. **Creativity** asks the learner to make or communicate mathematics. **Reflection** uses errors, proof, strategy and evidence. **Rest** protects re-entry and reduces simultaneous load. **Nutrition** is explicit in every lesson and identifies what is being nourished practically or intellectually.

Interactive entry and visual QA

- Every interactive has an explicit entry instruction, a stated noticing goal and a conceptual bridge.
- Check the complete visual construction, including attached shapes, labels, right-angle markers and target regions. The default state and intended interaction extremes must remain contained.
- In Movement activities ask: What can the learner change? What mathematical response can they observe?
- Could a learner opening this panel independently tell you what to do in the next 30 seconds? If not, the panel is not ready.

Scratchpad standard

Toolbar order: **Pen** -> **Line** -> **Colour** -> **Width** -> **Eraser** -> **Undo** -> **Redo** -> **Sticky Note** ->

Background. Mathematical backgrounds persist when learner ink is erased. Clear board removes learner marks and notes while preserving the selected background. Straight lines use the selected colour and width and participate in Undo/Redo and saved working state.

Safeguarding mathematical challenge

Support should reduce fog, not remove the mathematics. A learner may need one diagram, one step or one representation at a time. Preserve the central inquiry: what relationship is visible, why does it work, when may it be used, and what does the evidence justify?